

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

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Lab Assignment 11: Method Overriding

Experiment No. 11.1 - Shape and Square

Experiment Title:

Shape and Square

Aim:

To override area() in a Square class.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Method overriding means a child class provides its own version of a method that already exists in the parent class, with the same name and parameters. When the method is called on a child object, the child's version runs. A parent-type variable that holds a child object also runs the child's version; this is runtime polymorphism.

Algorithm / Procedure:

1. Define class Shape with area() returning 0.
2. Define Square extends Shape with field side and area() returning side * side.
3. Create a Square with side = 4.
4. Print Area with the result of area().

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 11, Task 1
// Question: Create class Shape with int area() returning 0, and
// class Square extending Shape with int field side and area()
// returning side times side. In main, create a Square, set side to
// 4, and print "Area:" followed by the value.

class Shape {
    int area() {
        return 0;
    }
}

class Square extends Shape {
    int side;

    @Override
```

```
    int area() {
        return side * side;
    }
}

public class Main {
    public static void main(String[] args) {
        Square s = new Square();
        s.side = 4;
        System.out.println("Area: " + s.area());
    }
}
```

Input:

Square side = 4.

Output:

```
Area: 16
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. Square.area() overrode Shape.area() and returned 16.

Viva-Voce Questions:

- What is method overriding?
- What does the @Override annotation do?
- What is the difference between overriding and overloading?

Experiment No. 11.2 - Shape and Rectangle

Experiment Title:

Shape and Rectangle

Aim:

To override area() in a Rectangle class.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Method overriding means a child class provides its own version of a method that already exists in the parent class, with the same name and parameters. When the method is called on a child object, the child's version runs. A parent-type variable that holds a child object also runs the child's version; this is runtime polymorphism.

Algorithm / Procedure:

1. Define class Shape with area() returning 0.
2. Define Rectangle extends Shape with fields length and breadth and area() returning length * breadth.
3. Create a Rectangle with length 4 and breadth 5.
4. Print Area with the result of area().

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 11, Task 2
// Question: Create class Shape with int area() returning 0, and
// class Rectangle extending Shape with int length and breadth,
// area() returning length times breadth. In main, create Rectangle
// with length 4 and breadth 5, and print "Area:" followed by the
// value.

class Shape {
    int area() {
        return 0;
    }
}

class Rectangle extends Shape {
    int length;
    int breadth;

    @Override
    int area() {
        return length * breadth;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Rectangle r = new Rectangle();
        r.length = 4;
        r.breadth = 5;
        System.out.println("Area: " + r.area());
    }
}
```

Input:

Rectangle length = 4, breadth = 5.

Output:

```
Area: 20
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. Rectangle.area() overrode Shape.area() and returned 20.

Viva-Voce Questions:

- Which version of area() runs for a Rectangle object?
- Can a child override a parent method with different parameters?
- What is the default area() of Shape?

Experiment No. 11.3 - Parent-type variable holding a Square

Experiment Title:

Parent-type variable holding a Square

Aim:

To demonstrate runtime polymorphism using a Shape variable that holds a Square object.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Method overriding means a child class provides its own version of a method that already exists in the parent class, with the same name and parameters. When the method is called on a child object, the child's version runs. A parent-type variable that holds a child object also runs the child's version; this is runtime polymorphism.

Algorithm / Procedure:

1. Define Shape and Square as before.
2. Create a Square with side = 3.
3. Store it in a variable of type Shape.
4. Call area() on the Shape variable and print the result.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 11, Task 3
// Question: Create class Shape with int area() returning 0, class
// Square extending Shape with int side, area() returning side times
// side. In main, create Square with side 3, store it in a Shape
// variable and print "Area:" followed by the value.

class Shape {
    int area() {
        return 0;
    }
}

class Square extends Shape {
    int side;

    @Override
    int area() {
        return side * side;
    }
}

public class Main {
    public static void main(String[] args) {
```

```
        Square sq = new Square();
        sq.side = 3;

        Shape s = sq;
        System.out.println("Area: " + s.area());
    }
}
```

Input:

Square side = 3.

Output:

```
Area: 9
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The Shape variable called Square's area() and printed 9, showing runtime polymorphism.

Viva-Voce Questions:

- What is polymorphism?
- Which area() runs when a Shape variable holds a Square?
- Why can a Shape variable not access the field side?